GEOPHYSICAL RESEARCH LETTERS, VOL. 21, NO. 6, PAGES 481-484, MARCH 15, 1994

Paleomagnetic Evidence for No Tectonic Rotation of the Central Italy Tyrrhenian Margin Since Upper Pliocene

Leonardo Sagnotti

Istituto Nazionale di Geofisica, Roma, Italy

Massimo Mattei, Claudio Faccenna and Renato Funiciello

Dipartimento di Scienze Geologiche, III Università di Roma, Roma, Italy

Abstract. We present paleomagnetic results from 141 samples collected at 12 sites from Plio-Pleistocene clayey units in the extensional Central Italy Tyrrhenian margin. All samples were subjected to stepwise alternating field and/or thermal demagnetization. Positive fold and reversal tests indicate that the results refer to a primary component of magnetization. Paleomagnetic mean direction (Declination = 358.0°; Inclination = 55.2°; α_{95} = 4.1°) reflects the expected geocentric axial dipole field for the locality, thus indicating that this sector of the Tyrrhenian margin did not undergo any tectonic rotation during the past 4 million years.

Introduction

The Italian peninsula underwent important geodynamic events during the Plio-Pleistocene, that caused extension and opening in the Tyrrhenian sea and coeval eastward migration and severe shortening in the Apenninic chain - Adriatic foredeep arcuate system. Paleomagnetic data collected during the past decade indicate that significant Plio-Pleistocene rotations are a common feature in the collisional environments of Southern Italy (Southern Apenninic chain, Calabrian Arc and Sicily foredeep) [Aifa et al., 1988; Sagnotti, 1992; Scheepers and Langereis, 1993; Scheepers et al., 1993]. No tectonic rotations have been found, however, for the Pleistocene in the Apulia foreland [Scheepers, 1992] and for the Upper Miocene - Pliocene in the extensional Tyrrhenian margin of Tuscany [Lowrie and Alvarez, 1979; Decini et al., 1990].

Most of the recent syntheses of the Central Mediterranean geodynamic evolution proposed a general counterclockwise rotation of the Italian peninsula since late Tortonian (8 Ma) in response to the opening in the Tyrrhenian sea (see i.e. Patacca et al., [1990]; Doglioni, [1991]; Castellarin et al., [1992]). Such reconstructions are based on the interpretation of regional geologic features and on the extrapolation, both in space and in time, of the paleomagnetic data from the mostly Meso-Cenozoic allochthonous units of the Apennines.

In order to investigate the relationships between the opening in the Tyrrhenian sea and the evolution of the Apennines, the neo-autochthonous Plio-Pleistocene clayey sediments of the Central Italy Tyrrhenian margin, that are suitable both for their tectonic position and age, were sampled for paleomagnetic analyses. The studied sector is characterized by the development of important fault-bounded transverse basins, interpreted as major transfer structures in the framework of the overall margin extension. Moreover, strike-slip shear zones in Pleistocene sediments were recently recognized [Faccenna et al., 1993]. These geologic features were investigated as a possible source of block-rotation mechanisms.

Copyright 1994 by the American Geophysical Union.

Paper number 94GL00393
0094-8534/94/94GL-00393$03.00

Geologic Setting

The structural setting of the Central Italy Tyrrhenian margin developed in a typical post-orogenic extensional regime and it is characterized by a system of NNW-SSE/NW-SE basins, filled by Upper Miocene to Lower Pleistocene syn-rift sediments (Figure 1). Coeval minor NE-SW basins developed as transverse structures. In the studied area such basins reach considerable dimensions. In particular, the Ardea basin is the most relevant, with a thickness of syn-rift and post-rift sediments up to 2500 m, bounded by normal faults with an half-graben geometry [Mariani and Prato, 1988]. Along the major faults roll-over structures occur. In the margin the extensional processes, that started in the Upper Miocene, progressively migrated eastward, following the compressional phases related to the build-up of the Apenninic chain [Elter et al., 1975; Patacca et al., 1990].

Upper Pleistocene N-S strike-slip faults were recently recognized in the Tiber valley north of Rome on the prolongation of a strike-slip system in the Apennines [Faccenna et al., 1993]. These *en-echelon* faults describe two narrow shear zones between the Sabina mountains and the Alban Hills volcano (Figure 1). Strike-slip tectonics was active up to late Pleistocene and locally gave rise to the development of small "pull-apart" basins, thermomineral springs, travertine deposition and volcano-tectonic features.

Crustal extension in the Central Tyrrhenian margin was accompained by volcanism, with volcanic centers following a NW-SE elongated sinking belt [Locardi et al., 1977]. Volcanic activity in the area started during the Pliocene (acidic series of the Tolfa-Ceriti area), but reached a climax in Middle-Upper Pleistocene with the onset of the huge volcanic complexes (high-potassium series) of the Roman Comagmatic Province.

Sampling and Measurements

Sampling was performed by *in situ* drilling using a diamond-head corer, cooled by water. 22 sites (Figure 1) were sampled in the grey-blue coloured fine-grained marine marly-clayey units, from different structural settings in the Tyrrhenian margin around Rome. The stratigraphic ages of the sampled units range from Upper Miocene (Messinian (6.5 - 5 Ma); sites LZ16 - Z17) to Lower Pleistocene (Emilian (1.4 - 1.1 Ma); site LZ02). The number of samples per site varies between 9 and 16, except at site LZ19, where we collected 47 samples in a 10 m thick

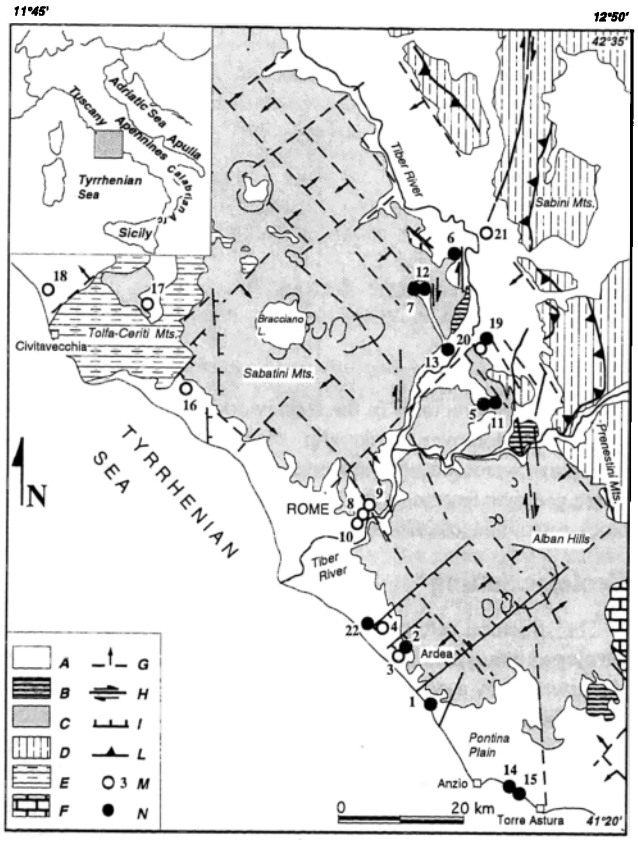


Figure 1. Geological sketch of the Central Italy Tyrrhenian margin.

Messinian to Recent units: A: sedimentary sequences of the "neo-autochthonous cycle" (Messinian - Pleistocene); B: travertine deposits (Quaternary); C: volcanic sequences (Pliocene - Pleistocene).

Meso - Cenozoic sedimentary sequences: D: Sabina units (Upper Triassic - Upper Miocene); E: Tolfa units (Upper Cretaceous - Eocene); F: Lepini Mts. units (Cretaceous).

Tectonic symbols: G: main buried faults, the arrow indicates the downthrow limb; H: strike-slip faults; I: normal faults; L: thrust faults.

Sampling sites: M: sites with NRMs too weak to be properly measured; N: sites measured and analyzed.

stratigraphic section for a magnetostratigraphic study. Five sites (LZ01-02-03-04-22) are in the Ardea basin, in tilted (up to 30°) sequences; two other sites (LZ14-15) were sampled along the coast just south of the Anzio village (Torre Astura). Nine sites (LZ05-06-07-11-12-13-19-20-21) are in the Tiber valley north of Rome, from almost sub-horizontal sequences affected by N-S to NE-SW right-lateral strike-slip faults. Three sites (LZ08-09-10) were sampled in the city of Rome (Monte Mario hill). Three more sites (LZ16-17-18) are in the Messinian-Pliocene basins of the Tolfa - Ceriti area.

Most of the paleomagnetic measurements were performed at the paleomagnetic laboratory of the Istituto Nazionale di Geofisica, on a JR-4 spinner magnetometer. Four sites (LZ01-06-09-16) were measured on a 2G cryogenic magnetometer at the laboratory of Gif-sur-Yvette (France). Natural Remanent Magnetization (NRM) values ranged between 20 and 50000 mA/m. For ten sites (LZ03-04-08-09-10-16-17-18-20-21) the NRM of the specimens was too weak to be accurately measured during demagnetization, even measuring on a cryogenic magnetometer. For the other twelve sites the NRM values were strong enough to be properly measured during the whole magnetic cleaning cycle. Two pilot specimens for each site were stepwise demagnetized, one by AF treatment, the other by thermal treatment, monitoring bulk susceptibility after each heating step. Increases in susceptibility occurred generally around 350 - 400°C. Results from each couple of pilots were generally consistent (Figure 2). For most of the sites the pilots indicated that the AF treatment gives a more efficient magnetic cleaning and a better defined demagnetization path. At site LZ19 some specimens carried a strong coercivity normal component of magnetization that disappears completely at 250 - 280°C, just before strong increases in bulk susceptibility. For this site combined treatment was used (thermal up to 310°C and then AF up to 80 mT). On the basis of the pilots behavior one specimen for each core was stepwise demagnetized with a Molspin AF demagnetizer (specimens treated at Gif-sur-Yvette were thermally demagnetized). Maximum peak fields varied, according to residual magnetization intensities, between 30 and 80 mT. Residual magnetizations at the end of the treatment were generally on the order of the 10% of the NRMs. Also, at each site anisotropy of magnetic susceptibility and structural analyses were performed. The details of this work and analysis of results will be published elsewhere [Sagnotti et al., 1993].

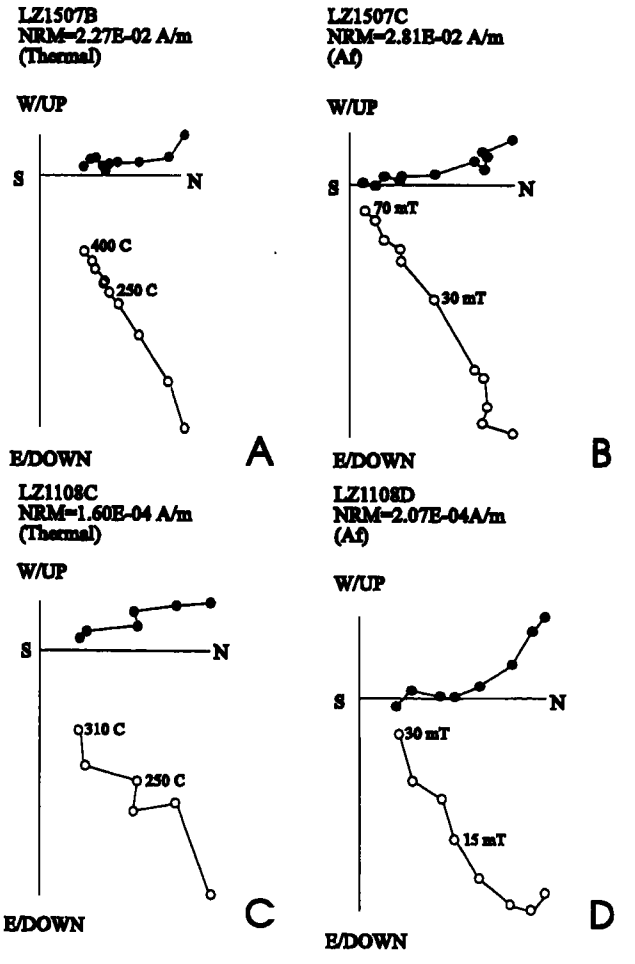


Figure 2. Orthogonal projection of demagnetization data for pilot specimens from sites LZ15 (A and B) and LZ11 (C and D). A and C: thermal demagnetization; B and D: alternating field demagnetization. Tilt corrected. Open circles - vertical projection; full circles - horizontal projection.

19448007, 1994, 6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94GL00393 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Results

For most of the measured specimens the demagnetization paths were well defined, giving either linear paths or remagnetization circles (Figures 3A and 3B). Best-fit lines and planes were evaluated by principal components analysis [Kirschvink, 1980] and accepted in the step range with the lower MAD (Maximum Angular Dispersion) value. Fits with MAD > 10° were disregarded. Site means were computed by combined analysis of linear paths and remagnetization circles, according to the method of McFadden and McElhinny [1988] (Figure 3C). Results are listed in Table 1. Dispersion of the data is generally very low, with an α_{95} parameter well below 10° for most of the sites. At seven sites it has been possible to apply a tectonic correction to the paleomagnetic data, restoring bedding to the horizontality. At the other sites bedding is horizontal. Five sites display a reversed polarity mean magnetization, the other seven sites a normal polarity mean magnetization. After tectonic correction, normal and reversed site mean directions group in two antipodal clusters (Figure 4; reversal test of type Rb according to McFadden and McElhinny [1990]: $\gamma_0 = 5.06°$; $\gamma_c = 8.44°$). Applying a tectonic correction to the seven sites with appreciable dip of the bedding, their site mean directions pass the fold test of McElhinny [1964] at the 99% confidence level ($k_2 / k_1 = 4.312$; $F_{(12,12)}1\% = 4.155$). The total mean of the 12 sites, after tectonic correction and transferring all data in the lower (normal polarity) hemisphere, is D = 358°, I = 55.2°, α_{95} = 4.1° (Figure 4 and Table 1).

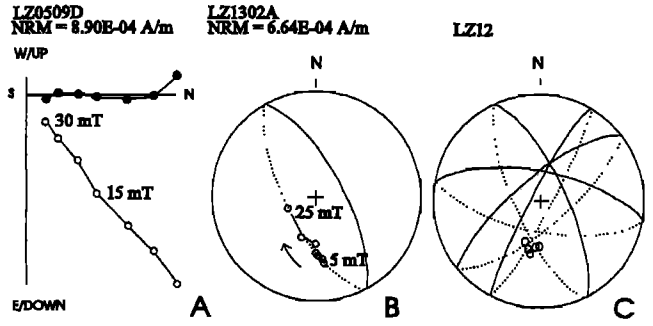


Figure 3. Representative demagnetization plots. Tilt corrected. A: Orthogonal projection of Af demagnetization of a specimen with a single stable component of magnetization. Open circle - vertical projection; full circles - horizontal projection. B: Equal-area projection of Af demagnetization of a specimen which defines a remagnetization circle. Solid line - lower hemisphere; open circles and dashed lines - upper hemisphere. C: Equal-area projection of linear paths (open circles) and remagnetization circles for site LZ12.

Conclusions

The paleomagnetic data are characterized by an overall low dispersion both at the site and the regional levels. The age of the measured sites is Upper Pliocene (Piacenzian; 3.5 - 1.8 Ma) for most of the sites and ranges between the uppermost Lower Pliocene (*G. puncticulata / G. margaritae* zone (4.2 - 3.5 Ma); site LZ22) to middle Lower Pleistocene (Emilian; site LZ02) (see Table 1). Positive fold and reversal tests indicate that the observed mean site directions reflect a primary component of magnetization. Thus, these data are suitable for the characterization of the geodynamic evolution of the area during the past 4 million years.

The total mean direction reflects with good approximation the expected geocentric axial dipole field direction for the locality (D = 0°; I = 60.5°). In fact, there is no significant vertical axis tectonic rotation recorded in the clayey units of the Central Italy Tyrrhenian margin, since uppermost Lower Pliocene. Flattening of the paleomagnetic inclination is of the order of few degrees.

Thus, we can argue that the strike-slip tectonics in the shear zone between the Sabina Mountains and the Alban Hills volcano was not able to activate paleomagnetically appreciable block-rotations. Also, the severe extensional, rifting and spreading processes that occurred in the Tyrrhenian sea during the past 4 million years (see i.e. Sartori, [1990] for a review) did not cause any tectonic rotation in the Central Tyrrhenian margin. We observe that the previously recognized Plio-Pleistocene rotations

Table 1. Summary of the Main Paleomagnetic Results

Site	Treat.	N(n,c)	D	I	A. Dip	Dip	D'	I'	k	α_{95}	Age*
LZ01	Th	7(0,7)	341.3	32.8	137	15	346.9	46.2	119.2	5.6	U. Pliocene
LZ02	Af	8(8,0)	158.6	-36.6	124	23	174.6	-53.8	161.3	4.4	L. Pleistocene
LZ05	Af	7(7,0)	352.6	51.2	99	6	0.1	52.5	150.1	4.9	U. Pliocene
LZ06	Th	7(5,2)	342.1	54.4	45	8	350.0	49.3	91.3	6.5	U. Pliocene
LZ07	Af	7(6,1)	189.1	-60.5	336	13	180.2	-49.0	151.5	5.0	U. Pliocene
LZ11	Af	10(5,5)	340.9	54.1	-	-	340.9	54.1	92.6	5.2	U. Pliocene
LZ12	Af	9(5,4)	175.9	-59.7	88	10	192.6	-58.6	206.7	3.7	U. Pliocene
LZ13	Af	9(1,8)	184.7	-58.2	-	-	184.7	-58.2	29.9	10.5	U. Pliocene
LZ14	Af	10(10,0)	5.7	64.4	-	-	5.7	64.4	182.5	3.6	U. Pliocene
LZ15	Af	10(10,0)	355.5	60.8	-	-	355.5	60.8	145.6	4.0	U. Pliocene
LZ19	Th+Af	47(33,14)	192.3	-53.0	282	6	184.4	-52.6	30.5	3.7	U. Pliocene
LZ22	Af	10(5,5)	7.1	58.6	-	-	7.1	58.6	28.0	9.6	L. Pliocene
Means:											
b.t.c. (7sites, 92 specimens)			351.4	50.5					34.3	10.5	
a.t.c. (7 sites, 92 specimens)							357.8	52.0	147.9	5.0	
Total (12 sites, 141 specimens)							358.0	55.2	115.1	4.1	

N: number of analyzed specimens (n=number of linear paths; c=number of remagnetization circles); Treat.: demagnetization treatment; D, I: declination and inclination, before tectonic correction; A. Dip: Azimuth of dip (bedding); Dip: Dip (bedding); D', I': declination and inclination, after tectonic correction; k: precision parameter; α_{95}: half-angle of the 95% confidence circle about the mean direction; b.t.c.: before tectonic correction ; a.t.c.: after tectonic correction; *: we refer to the stratigraphic scale and zonation in Patacca et al. [1990]. Means were computed transferring all the data in the lower (normal polarity) hemisphere.

19448007, 1994, 6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94GL00393 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

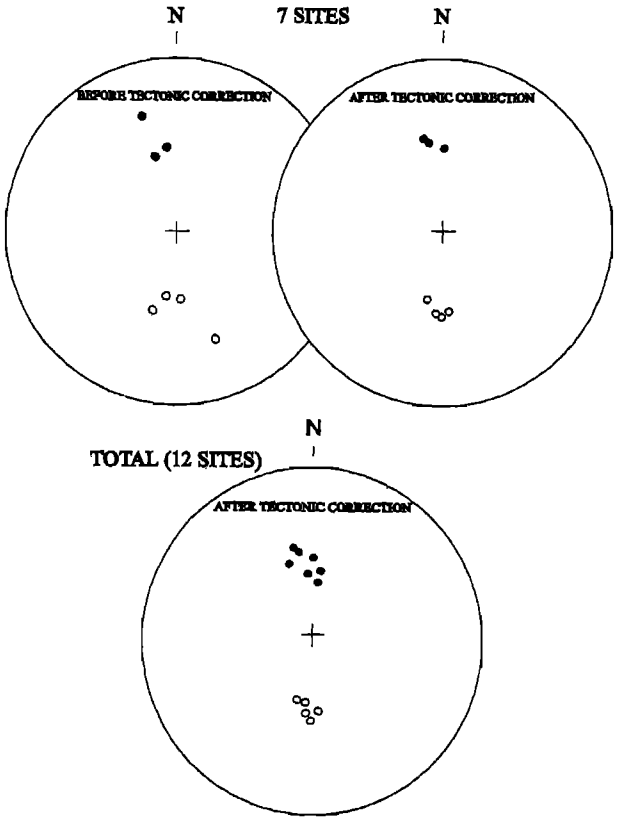


Figure 4. Equal-area projection of site mean directions from the 7 sites with appreciable tilting of the strata (above) and from all the measured sites (below).

in Southern Italy did not occur in the studied sector. Therefore, it is evident that, even for Plio-Pleistocene times, the Italian Peninsula cannot be considered as a unique homogeneously rotated "rigid block" and that the different geodynamic provinces, distinct on geologic bases, show difference also on their rotational pattern.

Acknowledgments. We wish to thank Fabio Florindo and Fabrizio Marra for their support during the field-work; Carlo Laj and Catherine Kissel, that allowed us to make measurements at their laboratory; Rob Van der Voo and an anonymous reviewer for helpfuls comments on the manuscript.

References

Aifa, T., H. Feinberg and J.P. Pozzi, Pliocene-Pleistocene evolution of the Tyrrhenian arc: paleomagnetic determination of uplift and rotational deformation, *Earth Planet. Sci. Lett.*, 87, 438-452, 1988.

Castellarin, A., C. Cantelli, A.M. Fesce, J.L. Mercier, V. Picotti, G.A. Pini, G. Prosser and L. Selli, Alpine compressional tectonics in the Southern Alps. Relationships with the N-Apennines. *Annales Tectonicae*, 6, 62-94, 1992.

Decini, D., R. Funiciello, M. Mattei, M. Tozzi, C. Kissel and C. Laj, Dati paleomagnetici preliminari nelle unità neogeniche del margine tirrenico tosco-laziale, *Mem. Soc. Geol. It.*, 45, 569-575, 1990.

Doglioni, D., A proposal for the kinematic modelling of W-dipping subductions - possible applications to the Tyrrhenian-Apennines system, *Terra Nova*, 3, 423-434, 1991.

Elter, P., G. Giglia, M. Tongiorgi, and L. Trevisan, Tensional and compressional areas in the recent (Tortonian to Present) evolution of north Apennines, *Boll. Geof. Teor. Appl.*, 17, 3-18, 1975.

Faccenna, C., R. Funiciello and M. Mattei, Late Pleistocene N-S strike-slip shear zone along the Latium Tyrrhenian margin: structural character and magmatological implication, in press on *Boll. Geof. Teor. Appl.*, 1993.

Kirschvink, J.L., The least-square line and plane and the analysis of paleomagnetic data, *Geophys. J. R. astr. Soc.*, 62, 699-718, 1980.

Locardi, E., G. Lombardi, R. Funiciello and M. Parotto, The main volcanic groups of Latium (Italy): relations between structural evolution and petrogenesis, *Geol. Rom.*, 15, 279-300, 1977.

Lowrie, W. and W. Alvarez, Paleomagnetism and Rock Magnetism of the Pliocene Rhyolite at San Vincenzo, Tuscany, Italy, *J. Geophys.*, 45, 417-432, 1979.

Mariani, M. and R. Prato, I bacini neogenici costieri del margine tirrenico: approccio sismico-stratigrafico, *Mem. Soc. Geol. It.*, 41, 519-531, 1988.

Patacca, E., R. Sartori and P. Scandone, Tyrrhenian Basin and Apenninic Arcs: kinematic relations since late Tortonian times, *Mem. Soc. Geol. It.*, 45, 425-451, 1990.

McElhinny, M.W., Statistical significance of the fold test in paleomagnetism, *Geophys. J. R. astr. Soc.*, 8, 338-340, 1964.

McFadden, P.L., and M.W. McElhinny, The combined analysis of remagnetization circles and direct observations in paleomagnetism, *Earth Planet. Sci. Lett.*, 87, 161-172, 1988.

McFadden, P.L., and M.W. McElhinny, Classification of the reversal test in paleomagnetism, *Geophys. J. Int.*, 103, 725-729, 1990.

Sagnotti, L., Paleomagnetic evidence for a Pleistocene counterclockwise rotation of the Sant'Arcangelo basin, *Geophys. Res. Lett.*, 19, 135-138, 1992.

Sagnotti, L., C. Faccenna, R. Funiciello and M. Mattei, Magnetic fabric and structural setting of Plio-Pleistocene clayey units in an extensional regime: the Tyrrhenian margin of Central Italy, in press on *J. Struct. Geol.*, 1993.

Sartori, R., The main results of ODP Leg 107 in the frame of Neogene to Recent geology of perityrrhenian areas, in Kastens, K.A., Mascle, J. et al., *Proc. ODP, Sci. Results*, 107, 715-730, 1990.

Scheepers, P., No tectonic rotation for the Apulia-Gargano foreland in the Pleistocene, *Geophys. Res. Lett.*, 19, 2275-2278, 1992.

Scheepers, P. and C.G. Langereis, Analysis of NRM directions from the Rossello composite: implications for the tectonic rotation of the Caltanissetta basin, Sicily, *Earth Planet. Sci. Lett.*, 119, 243-258, 1993.

Scheepers, P., C.G. Langereis and F.J. Hilgen, Counter-clockwise rotations in the southern Apennines during the Pleistocene: paleomagnetic evidence from the Matera area, *Tectonophysics*, 225, 379-410, 1993.

L. Sagnotti, Istituto Nazionale di Geofisica, Via di Vigna Murata 605, 00143, Roma, Italy (e-mail: sagnotti@in8800.cineca.it)

Massimo Mattei, Claudio Faccenna and Renato Funiciello, Dip. di Sc. Geologiche, III Università di Roma, Via C. Segre 2, 00154, Roma, Italy

(Received December 15, 1993; accepted December 26, 1993.)

19448007, 1994, 6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94GL00393 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License